

DSA0502 – Query Processing for Data Science	
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LAB PROGRAMS 17 AND 18

Program 17

Question: Split dataframe by school code and get mean, min and max age.

Aim: To implement the given Pandas program successfully.

Dataset :

School_Code	Class	Student	Age
S101	10	Asha	15
S101	10	Ravi	16
S102	9	John	14
S102	9	Mary	15

Code

```
import pandas as pd
df=pd.read_csv('school.csv')
print(df.groupby('School_Code')['Age'].agg(['mean','min','max']))
```

Output

Mean/min/max age.

	mean	min	max
School_Code			
S101	15.5	15	16
S102	14.5	14	15

Result

Thus, the program was executed successfully and the expected output was obtained.

Program 18

Question: Split dataframe into groups based on school code and class.

Aim: To implement the given Pandas program successfully.

Dataset :

School_Code	Class	Student	Age
S101	10	Asha	15
S101	10	Ravi	16
S102	9	John	14
S102	9	Mary	15

Code

```
import pandas as pd
df=pd.read_csv('school.csv')
for k,v in df.groupby(['School_Code','Class']):
    print(k)
    print(v)
```

Output

Groups by school and class.

```
... ('S101', np.int64(10))
   School_Code  Class Student  Age
0         S101     10   Asha   15
1         S101     10   Ravi   16
... ('S102', np.int64(9))
   School_Code  Class Student  Age
2         S102      9   John   14
3         S102      9   Mary   15
```

Result

Thus, the program was executed successfully and the expected output was obtained.